

Name: _____ **Date:** _____

Recall the following properties of derivatives:

(1) $\frac{d}{dx}(f(x) + g(x)) = \frac{df}{dx} + \frac{dg}{dx}$ for any functions $f(x), g(x)$.

(2) $\frac{d}{dx}(a \cdot f(x)) = a \cdot \frac{df}{dx}$ for any function $f(x)$ and any number a .

(3) $\frac{d}{dx}(x^n) = nx^{n-1}$ (we call this the **Power Rule**).

Problem 1. Use the power rule to find the derivatives of the following functions:

a) $f(x) = 3x - 5$

b) $f(x) = x^2 - 5x + 2$

c) $f(x) = 4x^3 - 6x + 10$

d) $f(x) = 8x^2 - 16x + 5$

e) $f(x) = x^4 - 4x^3 + 12x^2 - 5x + 1$

f) $f(x) = x^5 + x^4 + x^3 + x^2 + x + 1$

g) $f(x) = 6x^{10} + 5x^4 - 8x^2 + 100$

h) $f(x) = x^{100} - x^{50} + x^{25}$

Below is a list of some common derivatives:

- $\frac{d}{dx}(\sin(x)) = \cos(x)$
- $\frac{d}{dx}(\cos(x)) = -\sin(x)$
- $\frac{d}{dx}(\tan(x)) = \sec^2(x)$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
- $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

Problem 2. Use the above table of derivatives, as well as properties (1) and (2) from page 1, to find the derivatives of the following functions:

a) $f(x) = \sin(x) + \cos(x)$

b) $f(x) = 5e^x$

c) $f(x) = \tan(x) - \ln(x)$

d) $f(x) = 3\sqrt{x} + 4\cos(x) - 2\ln(x) + e^x$

e) $f(x) = -2\sin(x) - 3\tan(x) + \sqrt{x}$